

Interactive Paper Displays: Enabling Digital Functionalities on Physical Paper

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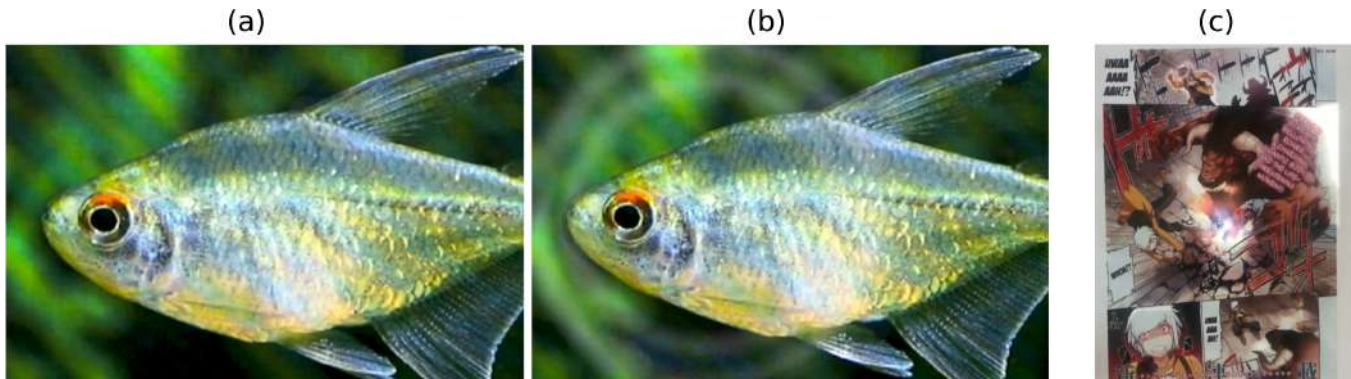


Figure 1: Examples of the *Pinhole Paper* and dynamic illumination. (a) An image printed with ink-jet printer on a perforated paper. (b) As soon as the display below the paper shows dynamic luminance animation of water, the user perceives it as a genuine water animation on the printed image. (c) A comic book on a perforated paper enhanced with dynamic illumination shown by the display below the paper.

ABSTRACT

Despite numerous possibilities to combine digital functionalities with physical paper, these solutions require users to handle, wear (e.g. smart phones, head-mounted displays) or set up (e.g. projectors) additional hardware, or require special manufacturing (e.g. embedding electronics, using conductive ink). We overcame these limitations by designing and developing several new interactive paper-display technologies that allow one to interact with paper and digital content, when paper is placed on a horizontal interactive display such as a table computer (tabletop) or tablet. Since the content on the display is combined with the content on paper,

the screen real-estate the paper is covering is not lost. We present twelve practical applications: various interactive learning exercises and drawing aids as well as instant digitisation of handwriting when filling a form on paper, annotating a blueprint or writing a sticky note. With these we demonstrate various new opportunities of adding digital functionalities to paper and aim to initiate a debate on the future role of paper in our society.

CCS CONCEPTS

• **Human-centered computing** → *Displays and imagers*.

KEYWORDS

paper interfaces, interactive paper, rear projection, pinhole paper, paper display, digitiser, apple pen

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1 INTRODUCTION

Despite the digital age we live in, paper still retains a significant role in our daily activities; be it in our working environments, classrooms or at home. Its long lasting existence is due to its unique properties such as simplicity of use, endurance and flexibility. Nevertheless, digital mediums also offer several advantages over paper, such as displaying multimedia content and being connected to the internet providing instant access to a vast amount of information. To this end it is not surprising that numerous attempts to combine the richness of paper interaction with the flexibility of digital mediums have been proposed and explored [6].

If we are to use regular (not electronically enhanced) paper and want to show digital content together with the content on paper (not beside it such as on an adjacent computer screen) one option is to use augmented reality (AR). However, AR applications demand users to handle or wear additional hardware (e.g. smartphones, HMDs, or projectors) [1, 8, 10]. This can be a nuisance when augmenting an activity that primarily happens on the paper (e.g. augmenting reading without disrupting the reading experience per se [9]), or when users are expected to interact with paper for longer periods of time, such as when studying. Furthermore, such AR applications can require elaborated setups of hardware around users (e.g. complex projector placement) [7]. In addition, users can also occlude the augmentation preventing the system from functioning as intended.

To overcome these limitations, we have designed and developed several new interactive paper-display technologies that allow one to interact with the paper and digital content, when the paper it is placed on a horizontal interactive display such as a table computer (tabletop) or tablet. These include: the (i) *Pinhole Paper* [4], (ii) *Dynamic pinhole paper* [3], (iii) *Back-print illumination* paper display [2], and (iv) *The Hybrid Stylus*, a multi-surface active stylus for interacting with and handwriting on paper, tabletop display or both simultaneously [5].

By enabling interactivity with paper that is placed on a horizontal interactive display, the paper becomes “the screen” on which digital and printed content are viewed, whilst the “computer device” maintains its computational and input capabilities (e.g. touch and stylus). This approach does not demand additional physical effort from users (wearing additional hardware as well as switching between paper and computer screen), keeping the focus of attention on the paper. In addition, in all our paper-display technologies we preserve all paper affordances and usage of tools such as pen, pencil or digital stylus. This provides several novel interaction possibilities to paper, which we plan to present within this interactivity.

2 PINHOLE PAPER: MAKING STATIC OBJECTS PERPETUALLY DYNAMIC

With the *Pinhole Paper* [4] we achieved an augmentation technique that modifies the content appearance on paper by creating an illusion of a static object appearing perceptually dynamic. We based our approach on the Deformation Lamps technique (DL) [7]. However, to make the illusion work, we modified: (i) the DL method of

generating dynamic luminance animation by adding gamma and brightness correction; and (ii) the optical and mechanical properties of paper by perforating it with a grid of 0.25 mm holes. The perforation decreases the opacity of the paper, allowing more light to pass through while preserving all other paper properties. This concept can be used to create new reading experiences by incorporating subtle reactive animations to illustrations in graphic novels or any book that contains visual imagery. For example, in an illustration of a fish swimming, the system can generate waves of water to create a more convincing viewing experience (see Figure 1b). Another example is presented in Figure 1c where a physical comic book is augmented with animated illumination of an explosion, creating dynamic effects to the physical content on paper. Other dynamic luminance animations can be added, which is limited only by the imagination of the authors. It also is important to note, that the content on paper can be consumed without such dynamic effects.

3 DYNAMIC PINHOLE PAPER: IMPROVING PERCEIVED QUALITY OF DIGITAL IMAGERY

In *Dynamic pinhole paper* [3] we combined the content on paper with the digital content beneath the paper when placed on a horizontal interactive display. However, placing displays beneath the paper affects the quality of the image on the “paper screen”. To improve it one can perforate paper with tiny holes (i.e. pinhole paper) enabling users to directly observe fragments of the underlying display and allow for seamless combination of physical and digital content on the paper itself, while preserving the interactive capabilities of the display as well as all paper qualities. Nevertheless, perforation also reduces the resolution of the digital content since it can be observed only where paper is perforated. To address this challenge we proposed to move digital content to reveal occluded areas by the paper. We have demonstrated that it is possible to increase the resolution of the digital content below the paper by moving the content on the screen at appropriate speed and on an appropriate trajectory [3].

To demonstrate various use-cases, we built several prototypes including: A geography exercise book showing hints through a perforated area of the paper when the user taps on a particular country (Figure 2a); A form offering hints for each field in multiple languages selected by tapping on a flag visualised through perforated area until the flag of the desired language is shown (Figure 2b); A mathematics exercise showing step-by-step guides where each step can be accessed by tapping on an associated note besides perforation (Figure 2c).

4 LIGHTMEUP: BACK-PRINT ILLUMINATION PAPER DISPLAY

Back-print Illumination Paper Display [2] is a concept of paper containing multi-stable visuals that are printed on the back of the paper. The back-printed content only becomes visible once paper is back-illuminated with a particular colour. Users can place a paper on an interactive screen and selectively reveal additional information by controlling back-light. In our case, the back-light is controlled via a combination of on screen gestures, spotlight technique, and see-through lenses (i.e. circular, square). We can



Figure 2: Prototypes of *Dynamic pinhole paper*: (a) A geography exercise book; (b) A form offering hints in multiple languages; (c) A math exercise showing step-by-step guides.

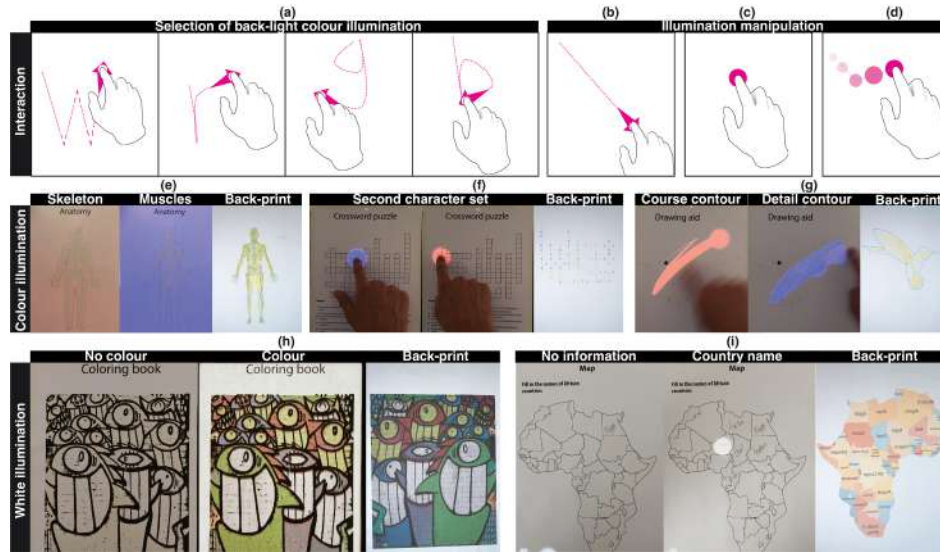


Figure 3: *Back-print illumination* interaction techniques: (a) Colour selection, (b) Fullscreen illumination, (c) Spotlight illumination and (d) Chasing spotlight illumination. *Back-print illumination* prototypes: (e) Anatomy book, (f) Crossword puzzle, (g) Drawing aid, (h) Colouring book and (i) Geography exercise book. For a better illustration of these use-cases see the accompanying video figure.

categories these interaction techniques into two groups: the selection of back-light colour (see Figure 3a), and the illumination manipulation (see Figure 3b). *Back-print illumination* techniques open up several interesting use cases described in detail below.

Drawing aid. The drawing aid use-case depicted in Figure 3g shows only initial instructions available on the front side of the paper. However, if users need more help, they can decide to either reveal the outline of the bird, the inner details of the bird, or both by selecting the appropriate back-light colour on the screen below. Such multi-stable visuals enable one to provide different levels of support (e.g. outline vs. inner detail support) as well as allow users to selectively reveal only areas where help is needed (chasing spotlight illumination) or display all support at once (fullscreen illumination). Another example of a drawing aid are colouring books (Figure 3h).

To help users in selecting appropriate colours (e.g. colours that form a matching colour palette) white back-illumination can reveal hints on how to colour a particular contour. As before, spotlight illumination can reveal only parts of the back-printed information, while fullscreen illumination can reveal all content on the back of the paper.

Exercise books. Exercise books are an important learning tool. Our first use case demonstrates an anatomy exercise book that shows general context as a contour drawing on the front side of the paper (Figure 3e), whereas the other back-printed visuals can be used to selectively reveal additional information (i.e. skeleton, nerves and vascular system). Another example of multi-level support is a crossword puzzle (Figure 3f), which reveals a different set of characters (i.e. consonants and vowels) using different back-light

colour illumination. The last example presented is a geography exercise book, which shows a contour drawing of the Africa continent on the front side of the paper (Figure 3i). The back-printed information can reveal additional content or hints within each country border, such as the name of the country, capital city, population to name a few. A similar approach can be adapted for other school subjects such as mathematics, biology, etc. As with previously described *Dynamic pinhole paper*, one of the main benefits of this approach is that the information is shown on paper in context where the learning is taking place. This minimises the need of searching for hints and help on other mediums and consequently minimises interruptions during the learning process.

5 THE HYBRID STYLUS

The *Hybrid Stylus* [5] is an extended active stylus that enables one to simultaneously digitise handwritten information on the paper when it is placed on a top of a touchscreen. To achieve this, we designed an extension consisting of an infinity pencil tip made of graphite that is attached to an active stylus. The tip acts as a normal pencil whilst maintaining all conductive properties of an active stylus. The *Hybrid Stylus* thus presents a low-cost alternative to handwriting digitisation using modified off-the-shelf hardware. Furthermore, the *Hybrid Stylus* can accompany and enhance all previously described concepts as it enables a secondary input method and supports a simple and precise paper alignment strategy. We designed and developed three use-case prototypes that demonstrate the *Hybrid Stylus* possibilities.

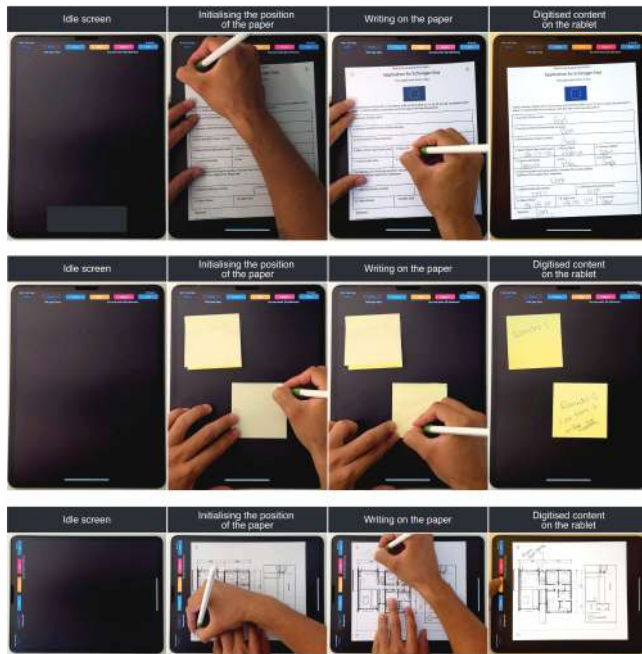


Figure 4: Hybrid Stylus prototypes. Top: Form filling prototype. Middle: Writing sticky notes. Bottom: Annotating blueprints.

Form filling. Forms, such as visa applications, medical diagnoses, or check lists are common in all segments of our society.

Users can place the form on the tablet screen (Figure 4 top) and align its position by performing stroke actions over printed target markers. When the position is aligned, the application places a digital counterpart of the physical form exactly below the paper. By doing so we enable users to input the required information synchronously on paper and the tablet computer.

Writing sticky notes. Using sticky notes is a popular way to quickly externalise ephemeral information such as writing down reminders. However, users tend to lose, misplace or misshape such notes so information can get lost. Our prototype (Figure 4 middle) enables users to simultaneously generate digital and physical sticky notes. As such, physical sticky notes maintain their original affordances whilst the digital copy makes it more difficult to lose such information.

Annotating blueprints. At construction sites the architects and builders need to agree on solutions to problems that arise during construction. This usually requires immediate action and happens in the context of a building site by drawing solutions and annotations on paper blueprints. However, these blueprints are held at the construction site, which makes it difficult to synchronise digital versions of the blueprints with the new up-to-date information added on paper. Blueprint annotation prototype attempts to resolve this problem by enabling the architects and builders to annotate the blueprints by sketching and writing directly on the blueprint whilst the annotations are simultaneously also digitised. After the annotations are made, the hard copy can be given to builders whilst the architects keep the digital counterpart (Figure 4 bottom).

6 CONCLUSION

In this interactivity we presented four new interactive paper display concepts that bring various digital functionalities to physical paper. By demonstrating twelve practical applications we show various novel use-cases of such paper-based technologies. Our prototypes provide the basis to several novel ways of expanding current paper interaction and blending physical with digital content seamlessly on the paper. This can instigate the debate on the future role of paper in the digital world.

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